Genetic Sequencing Algorithms Allow for the Discovery of Directed Patterns in Patient Sharing Pathways

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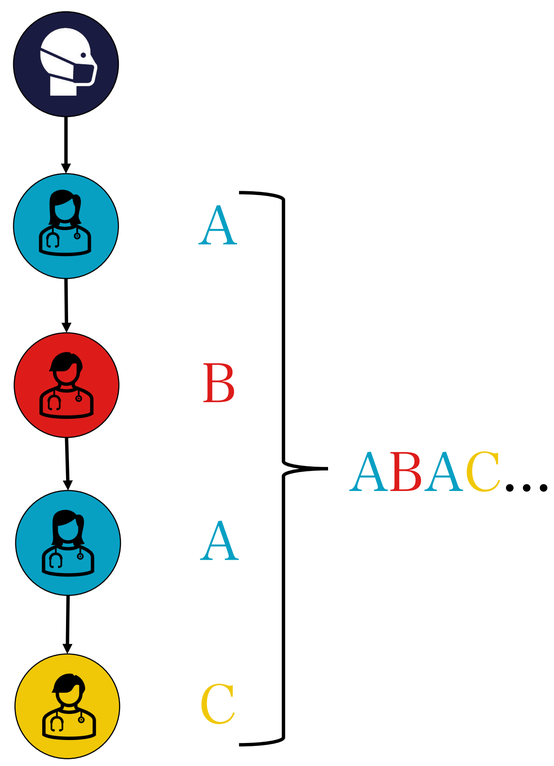
Patient Sharing Pathways



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ABCDCDEDCB
ABACACDEFAGFAGAGHAFAGABAI
ABCABCACACADCAD
ABACDCBCBCEFB
ABABABABABAB
ABACADBABCEBCBACADBEADCFB
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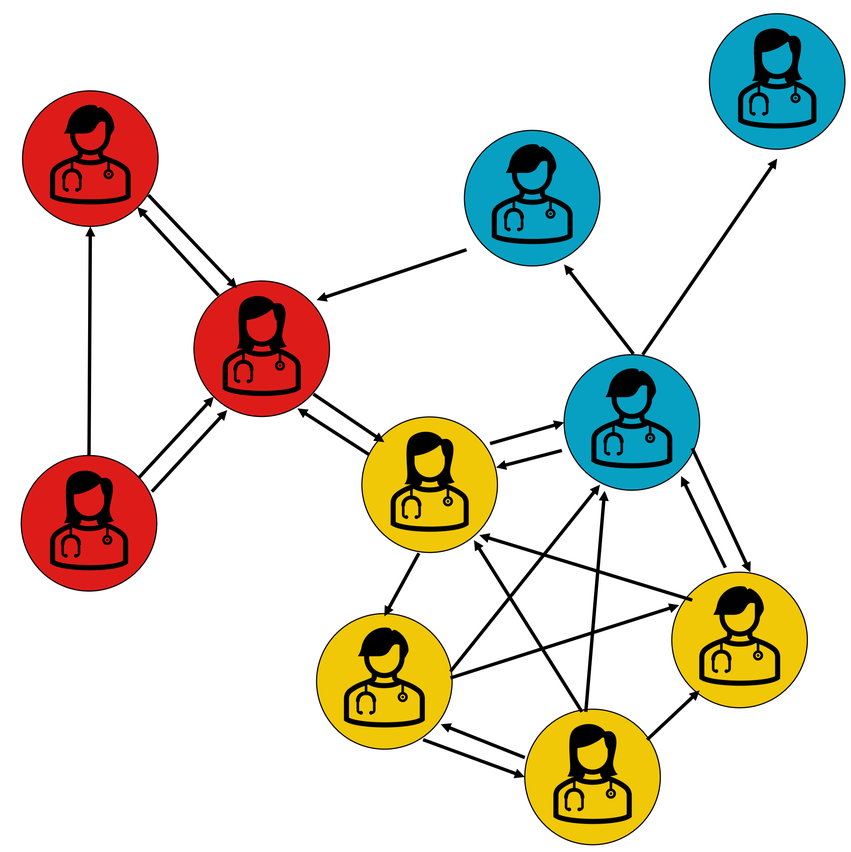
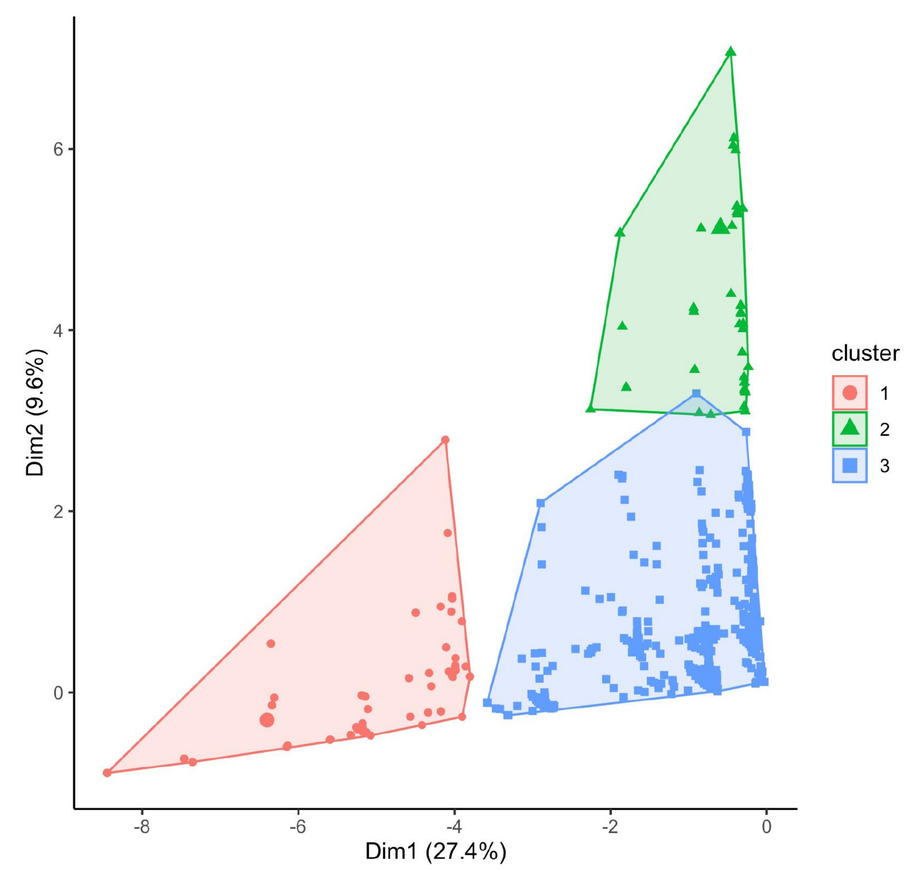
Patient Sharing Networks



Methods

- Patient sharing pathways were constructed from Medicare data
- We restricted to 1000 randomly selected patients who had at least 10 Medicare claims in 2018
- Pathways were encoded such that the first physician is labelled 'A', the second 'B', etc.
- Sequences were analyzed using *Biostrings*[1] and *universalmotif*[2]
- Counts of kmers in each patient path were used for kmeans[3] clustering

kmer	count
ABA	1255
BAB	909
ABAB	796
ABC	712
BABA	663
ABABA	607
BCB	511
BABAB	483
ACA	414
CBC	364
BCD	351
BAC	342
CAC	319
BCBC	295
ABAC	289



Discussion

- Patient sharing networks study the physician ecosystem on observable outcomes
- Methods studying 'omics networks may provide insight in studying physician groups
- This analysis supports the hypothesis that physician revisits in patient pathways occur frequently and should be considered in the optimization of patient sharing networks

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Works Cited

1. H. Pagès, P. Aboyoun, R. Gentleman and S. DebRoy (2021). Biostrings: Efficient manipulation of biological strings. R package version 2.60.2. https://bioconductor.org/packages/BiostringsBenjamin
2. Jean-Marie Tremblay (2021). universalmotif: Import, Modify, and Export Motifs with R. R package version 1.10.2. https://bioconductor.org/packages/universalmotif/
3. Lloyd, S. P. (1957). Least squares quantization in PCM. Technical Report RR-5497, Bell Lab, September 1957.